



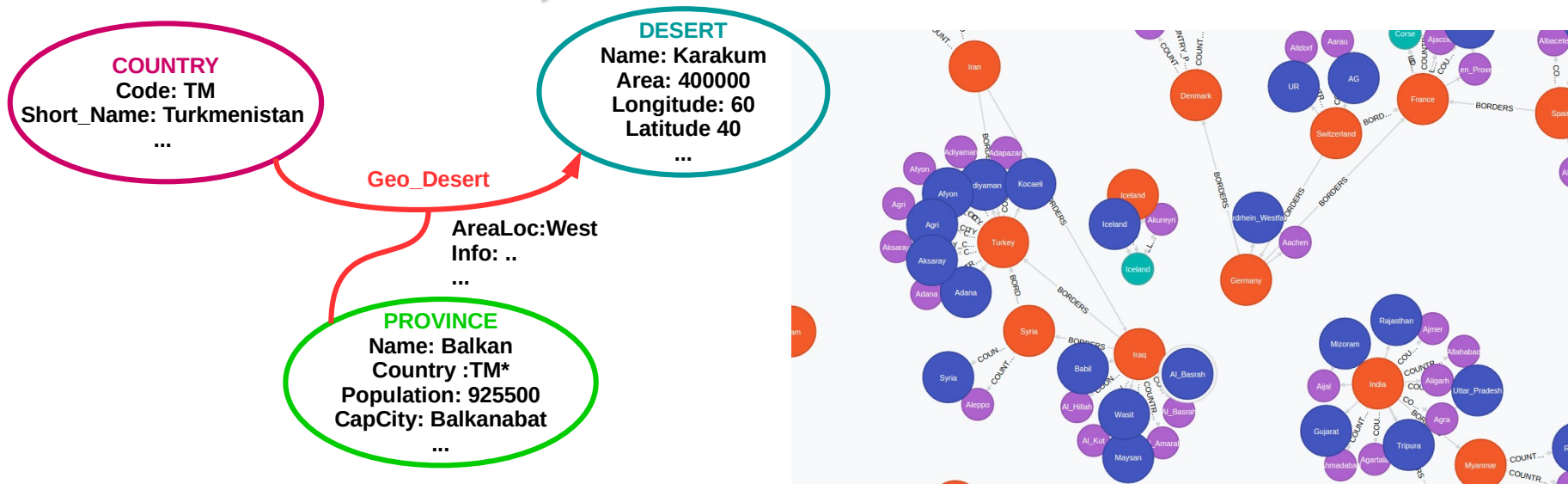
An Attributed Predicate-RDF Knowledge Base and Reasoning

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Introduction (1/2)

- Graph models have some advantages and are more expressive in representing data ^[1]. The models have inspired an emerging database technology, named graph databases ^[2].



1. C. C. Aggarwal and H. Wang, Managing and mining graph data, vol. 40. Springer, 2010.
2. R. Hecht and S. Jablonski, "NoSQL evaluation: A use case oriented survey," 2011.



Introduction (2/2)

- The data in Semantic web technology is a connected data, a linked data ^[3].
- An RDF ^[4] as it's data model must represent any knowledge of the real world.
- Native RDF stores have some limitations in representing a graph oriented of the data ^[5].

:Turkey :hasData :AG_AGR_TRAC_N
(S) (P) (O)

3. T. Berners-Lee, J. Hendler, O. Lassila, and others, "The semantic web," Sci. Am., vol. 284, no. 5, pp. 28–37, 2001.

4. W. W. W. Consortium and others, "RDF 1.1 Primer," 2014.

5. D. De Abreu, A. Flores, G. Palma, V. Pestana, J. Pinero, J. Queipo, J. Sánchez, and M.-E. Vidal, "Choosing Between Graph Databases and RDF Engines for Consuming and Mining Linked Data," in COLD, 2013.



Motivation

- Though simple, RDF is the standard data model of Semantic Web technology ^[6].
- Intuitively, an RDF is a graph-like ^[7]. In our opinion, RDF can manage graph oriented data as well, including the complex one, property hypergraphs.

:Turkey :hasData :AG_AGR_TRAC_N
(S) (P) (O)



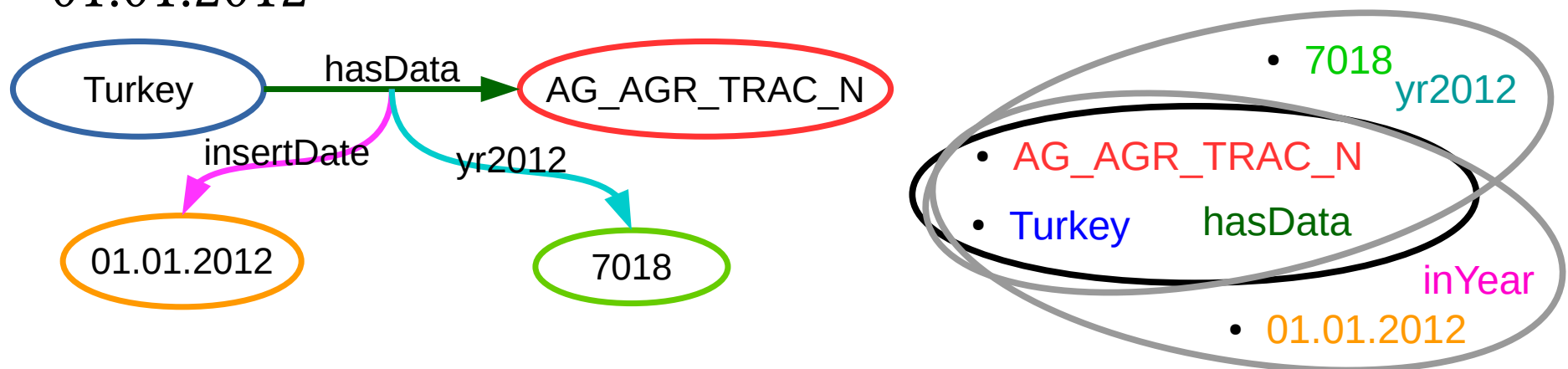
6. F. Manola, E. Miller, and B. McBride, "RDF primer," W3C Recomm., vol. 10, 2004.

7. J. Hayes and C. Gutierrez, "Bipartite Graphs as Intermediate Model for RDF," in The Semantic Web – ISWC 2004, vol. 3298, S. McIlraith, D. Plexousakis, and F. van Harmelen, Eds. Springer Berlin Heidelberg, 2004, pp. 47–61.



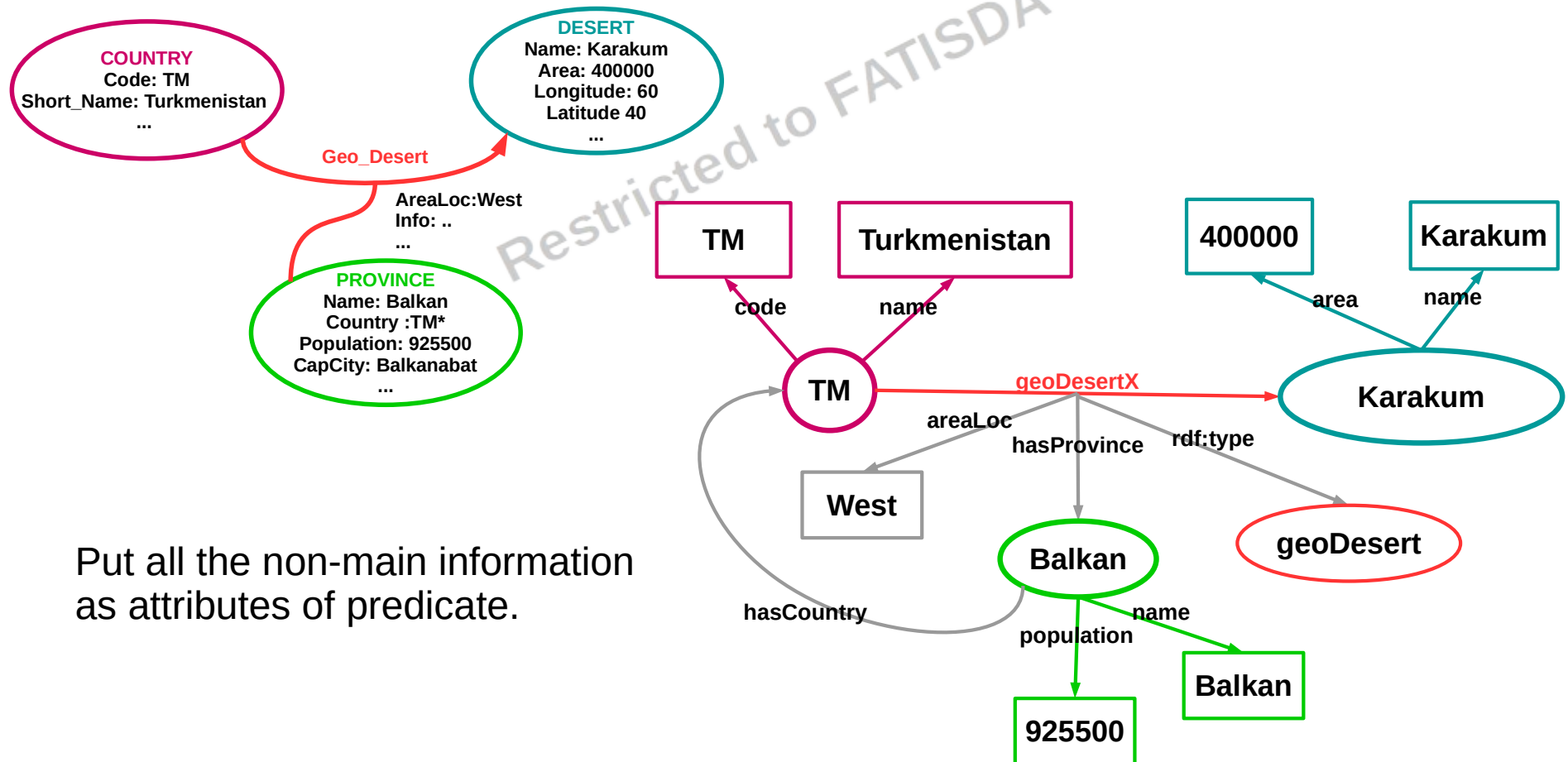
An AP-RDF (1/3)

- Predicate to manage all additional information of binary data.
- The subject and object are closed to be elaborated, but the predicates are still open ^[7].
- Ex: Turkey has data indicator AG_AGR_TRAC_N, the number for year 2012 is 7018 and was inserted on 01.01.2012





An AP-RDF (2/3)



Put all the non-main information
as attributes of predicate.



An AP-RDF (3/3)

@prefix : <http://www.semwebtech.org/ex/10/meta#> .

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .

@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

:TM :code "TM"^^xsd:string .

:TM :name "Turkmenistan"^^xsd:string .

:Karakum :name "Karakum"^^xsd:string .

:Karakum :area "400 000"^^xsd:float .

:TM :geoDesertX :Karakum .

:geoDesertX rdf:type :geoDesert .

:geoDesertX :areaLoc "West"^^xsd:string .

:geoDesertX :hasProvince :Balkan .

:Balkan :name "Balkan"^^xsd:string .

:Balkan :country :TM .

:Balkan :population "925 500"^^xsd:float .

The obtained AP-RDF (.ttl)